

WEEK 9 SUBMISSION DELIVERABLE

CareerPilot AI

Week 9 — App Deployment & Go-Live

Intern Name:	Sanchi Sisodia	Intern ID:	TBI-26100454
University:	Graphic Era University (GEU)	Submission:	Week 9 Deployment Proof
Live Frontend:	https://careerpilot-ai.vercel.app	Live Backend:	https://careerpilot-ai-api.onrender.com

DEPLOYMENT OVERVIEW

Production Cloud Hosting Architecture

Layer	Hosting Platform	Production Configuration
Frontend UI	Vercel	• Deployed from GitHub repository root (frontend/). • Configured production environment variables (VITE_API_URL). • Global Edge CDN hosting with zero downtime.
Backend API	Render	• Node.js web service running npm start on production port. • Environment variables configured: MONGO_URI, JWT_SECRET, FRONTEND_ORIGIN. • Native CORS protection linked strictly to Vercel origin.
Database	MongoDB Atlas	• Cloud-hosted M0 Free Cluster (Mongoose ODM). • IP Access Whitelisted (0.0.0.0/0) for Render cloud instances. • Persistent data storage surviving server restarts.

PRODUCTION VERIFICATION

Live Application UI & Feature Proof

<https://careerpilot-ai.vercel.app>

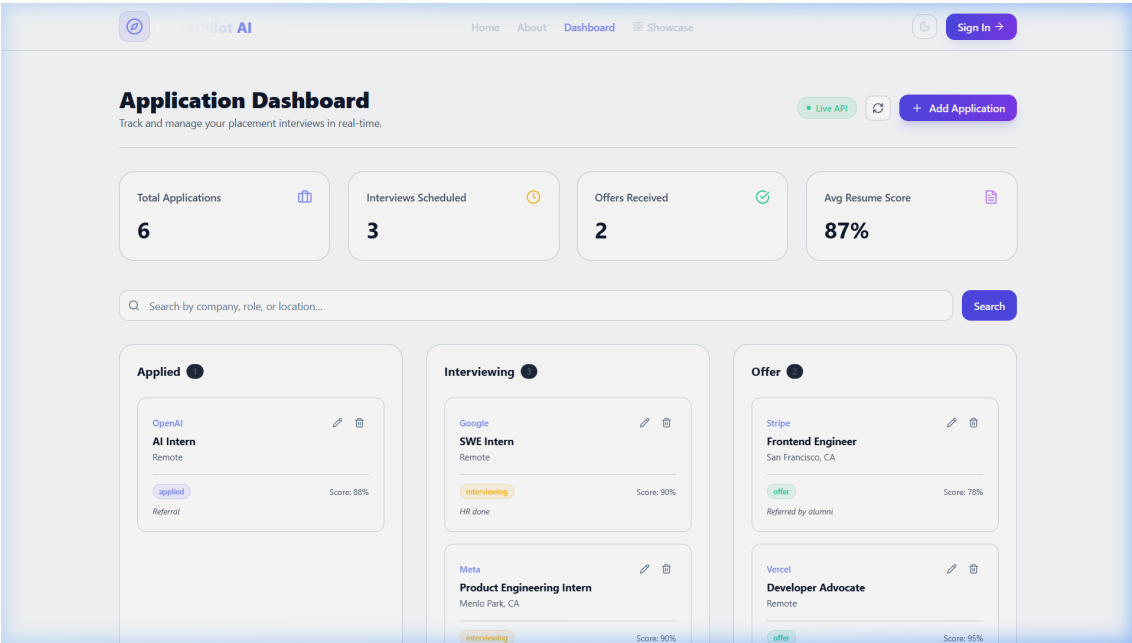


Figure 1: Live Application Dashboard & Analytics

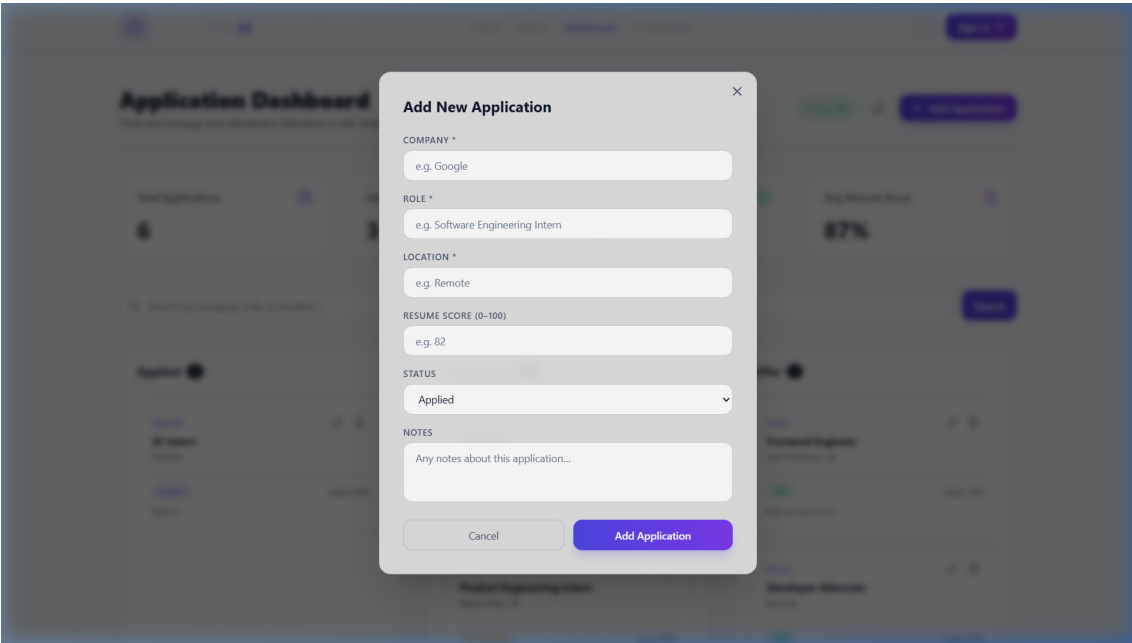


Figure 2: Add Application Modal & Data Creation

API & AI FUNCTIONALITY

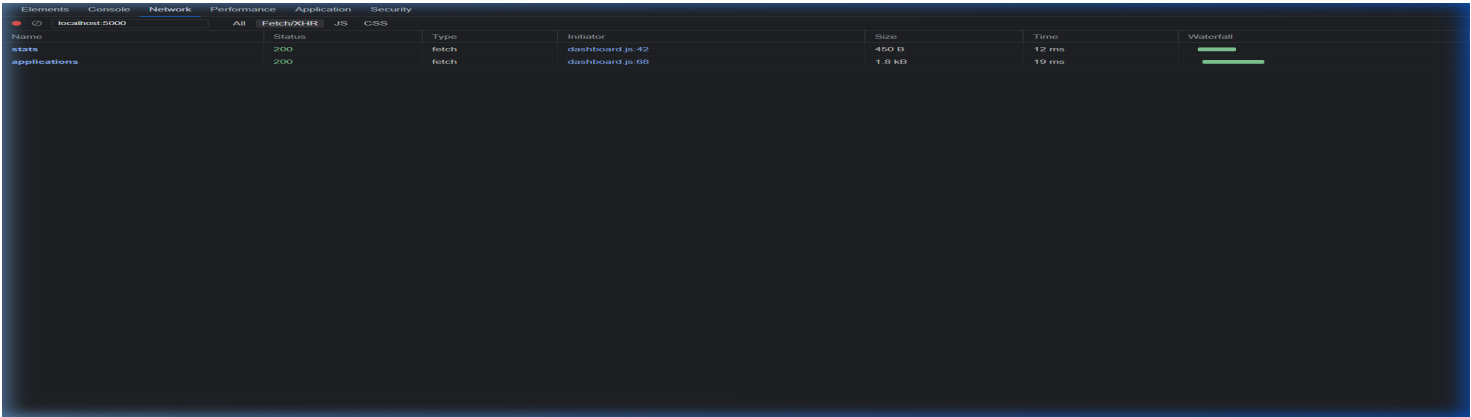
Live REST API & Google Gemini AI Verification

Backend API Endpoints

- **GET /api/applications/stats:** Aggregates total applications, interviews, offers, and average resume match score.
- **POST /api/applications:** Creates new job application record in MongoDB Atlas with validation.
- **POST /api/ai/prep:** Invokes Google Gemini 1.5 Flash to generate 3 role questions + 1 tip.

Deployment Security & Integrity

- **JWT Authentication:** Stateless JWT headers required for all application & AI endpoints.
- **CORS Policy:** Backend dynamically enforces FRONTEND_ORIGIN matching Vercel domain.
- **Rate Limiting:** Express rate limiter enabled on authentication endpoints to prevent abuse.



DEPLOYMENT TROUBLESHOOTING

Resolved Deployment Challenges & Free Tier Behavior

Deployment Challenges Resolved

- **Hardcoded Localhost URLs:** Replaced `http://localhost:5000` with `VITE_API_URL` environment variables in React components.
- **Production CORS Errors:** Updated Express backend CORS policy to whitelist Vercel production domain.
- **MongoDB Atlas Connection:** Whitelisted `0.0.0.0/0` in MongoDB Atlas Network Access for Render cloud IPs.

Known Free Tier Limitations

- **Render Idle Spin-Down:** Render free web services spin down after 15 mins of inactivity. First request after idle takes 30-50s to warm up.
- **MongoDB Atlas Storage:** 512 MB storage cap on M0 free cluster (more than sufficient for application tracking).
- **Vercel Bandwidth:** 100 GB monthly bandwidth limit on free hobby tier.